

An Hourly Forecasting Framework for Urban Air Quality Classification in Cairo Using Hybrid Meteorological and Spatiotemporal Features

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Abstract—Cairo is a dense metropolitan area with high traffic volumes and industrial emissions, creating significant public health challenges due to ambient air pollution. To assist municipal decision-making, this study introduces a supervised machine learning framework to classify hourly air quality into three regulatory bands: “Good,” “Moderate,” and “Poor.” We utilize an extensive dataset of 72,312 continuous records from 2018 to 2026 calibrated precisely to Cairo local time. The predictive features include surface meteorological variables alongside key pollutant indicators (PM_{10} , $PM_{2.5}$, and NO_2). To capture local anthropogenic habits, a custom binary spatiotemporal feature representing the unique Friday-Saturday Egyptian weekend structure was engineered. Severe dataset class imbalance was handled strictly within the training partition using the Synthetic Minority Over-sampling Technique (SMOTE). We systematically evaluate five standalone classifiers—Logistic Regression (LR), k -Nearest Neighbors (k -NN), Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF)—and implement a heterogeneous Stacking Ensemble to combine their predictive strengths. Empirical validation shows that while basic linear configurations exhibit structural constraints due to non-linear atmospheric boundaries, ensemble learning markedly enhances performance. The Final Stacking Ensemble achieved a peak testing accuracy of 90.42%, establishing a highly dependable computational framework for real-time urban environmental monitoring.

Index Terms—Air Quality Index, Cairo, Machine Learning, Stacking Ensemble, SMOTE, Meteorological Features, Spatiotemporal Proxy

I. INTRODUCTION

Air pollution represents a severe environmental health hazard globally, particularly within crowded, semi-arid urban centers. Cairo, the capital city of Egypt, faces distinct atmospheric challenges caused by dense automobile transit networks, regional manufacturing output, and seasonal desert dust interactions. These factors result in highly variable concentration

patterns of fine particulate matter ($PM_{2.5}$), coarse particles (PM_{10}), and nitrogen dioxide (NO_2). Prolonged exposure to elevated levels of these criteria pollutants is linked to serious cardiovascular and respiratory issues. Developing robust data-driven infrastructures capable of mapping complex meteorological factors into real-time Air Quality Index (AQI) bands is therefore critical for municipal planning and healthcare readiness.

Traditional physical transport models require heavy mathematical formulations and significant computational power, often struggling to adjust when environmental conditions shift rapidly. In contrast, supervised machine learning (ML) paradigms offer a practical alternative by discovering complex, non-linear relationships directly from historical data. Surface weather parameters, including temperature variations, relative humidity, total precipitation, and wind speeds, deeply influence how pollutants suspend or disperse in the lower atmosphere. Concurrently, human activities follow strict cyclical timelines across the working week, which directly updates the baseline emission profile of the city.

This research implements an operational machine learning pipeline to categorize metropolitan air quality levels. By aligning multi-year hourly weather measurements fetched for Cairo local time with engineered temporal calendars, we contrast individual linear, distance-based, tree-based, and ensemble machine learning architectures. Through comparative empirical analysis, this paper highlights the benefits of using a stacking ensemble configuration to capture complex atmospheric changes, providing a reproducible and highly accurate blueprint for real-time urban environmental intelligence.

II. LITERATURE REVIEW

The application of machine learning for ambient air quality evaluation has grown significantly, shifting from traditional individual models toward modern heterogeneous ensemble configurations. Researchers globally leverage combinations of weather metrics and pollutant monitoring stations to forecast regional air quality bands.

In a multi-city environmental evaluation, a study explicitly analyzing air pollution characteristics in Cairo alongside other global capitals [1] demonstrated that traditional index approaches frequently overlook the synergistic health impacts of multi-pollutant mixtures. Their work notes that major metropolitan areas like Cairo exhibit dense, high-variance pollution profiles that require flexible, non-linear machine learning modeling to capture complex atmospheric and atmospheric dispersion interactions accurately.

To understand the operational boundaries of different algorithms, a comprehensive benchmark study published in Nature PMC Portfolio [2] systematically tested standard classifiers on large-scale air quality indexes. Their experimental results showed that traditional baseline models, such as standard linear estimators and basic Support Vector Machines, face clear limitations due to rigid decision surfaces when modeling volatile ambient variables. In contrast, tree-based frameworks like Random Forest adapted much better to non-linear data structures, proving that hierarchical tree aggregations handle data variance and localized atmospheric fluctuations more effectively than single-tree or straight-line models.

To address the performance ceilings of individual models, recent research has focused on combining diverse algorithms. A study introducing a generalized stacking architecture for air quality prediction [3] confirmed that relying on a single classifier family often results in weak generalization because individual models cannot capture all underlying patterns in volatile atmospheric data. By feeding the cross-validated probability outputs of different algorithm families into a secondary meta-learner, their stacking configuration significantly minimized individual model bias and variance, setting a highly robust standard for urban predictive pipelines.

Building on these published academic insights, our study applies a structurally diverse model stack (LR, k -NN, SVM, DT, and RF) to a localized, multi-year hourly dataset built specifically around Cairo’s weather and transit dynamics, incorporating severe imbalance correction and localized work-force calendars.

III. METHODOLOGY

A. Data Collection

The primary data stream was collected by targeting the geographical coordinates of metropolitan Cairo (30.0444°N, 31.2357°E) for an unbroken multi-year sequence from January 1, 2018, to April 1, 2026. Hourly concentrations of criteria pollutants were gathered from the Open-Meteo Air Quality API, isolating records for coarse particulate matter (PM₁₀), fine particulate matter (PM_{2.5}), and nitrogen dioxide

(NO₂). Simultaneously, continuous physical weather data was extracted from the Open-Meteo Archive API, obtaining hourly air temperature at 2 meters elevation (°C), relative humidity (%), wind speed at 10 meters elevation (km/h), and total liquid precipitation (mm). These streams were merged using their hourly timestamps, forming a complete database of 72,312 individual rows.

B. Data Preprocessing & Cleaning

An initial structural verification showed that the gathered dataset was highly complete, with a starting configuration of exactly (72312, 8) features. A missing value check confirmed zero null entries across all meteorological and chemical feature columns, indicating a continuous data stream. Duplicate checks were performed using the `drop_duplicates()` operation, verifying that no identical row-wise data vectors were present, which successfully prevented data leakage prior to partitioning.

C. Feature Engineering

To construct a supervised classification problem, a categorical target metric, `AQI_Class`, was derived from continuous PM_{2.5} values by mapping them to the standard United States Environmental Protection Agency (US EPA) boundaries:

$$\text{AQI_Class} = \begin{cases} \text{Good,} & \text{if } \text{PM}_{2.5} \leq 12.0 \mu\text{g/m}^3 \\ \text{Moderate,} & \text{if } 12.0 < \text{PM}_{2.5} \leq 35.4 \mu\text{g/m}^3 \\ \text{Poor,} & \text{if } \text{PM}_{2.5} > 35.4 \mu\text{g/m}^3 \end{cases} \quad (1)$$

This mapping produced an uneven class distribution containing 52,795 “Moderate” records, 17,162 “Poor” records, and 2,355 “Good” records.

D. Predictive Modeling & Classifier Architecture

The prepared mathematical features were trained across a diverse set of machine learning architectures to contrast their operational styles:

- **Logistic Regression (LR):** A baseline linear estimator that calculates multi-class probabilities using multinomial Softmax operations.
- **k -Nearest Neighbors (k -NN):** A non-parametric model that classifies points via majority voting within a localized Euclidean feature space.
- **Support Vector Machine (SVM):** Draws separating hyperplanes in high-dimensional space, utilizing kernel transformations to model non-linear boundaries.
- **Decision Tree (DT):** A hierarchical splitter that partitions data records step-by-step based on information gain rules.
- **Random Forest (RF):** An ensemble of independent decision trees built on random data subsets that produces predictions through averaged voting.
- **Final Stacking Ensemble:** A heterogeneous ensemble framework that takes the cross-validated probability predictions of the individual base models (LR, k -NN, SVM, DT, RF) and feeds them as input features into a final meta-classifier model to generate the ultimate prediction.

Before training, the dataset was split into training and testing sets using an 80/20 stratified split layout to perfectly preserve the underlying target class proportions. To handle the severe class imbalance among the target labels—specifically the small minority representation of the “Good” air quality category (2,355 records)—the Synthetic Minority Over-sampling Technique (SMOTE) was applied. This over-sampling step was executed exclusively on the training partition to prevent any risk of data leakage. By synthesizing new, non-replicated data points along the vector line segments joining neighboring minority instances, SMOTE balanced the class distribution so the models could learn minor class traits without majority-class bias. Following this over-sampling step, standard feature scaling via *StandardScaler* was applied to normalize feature variances across the data vectors for distance-sensitive configurations.

IV. EMPIRICAL RESULTS

The performance outcomes from our machine learning experiments are recorded in Table I, displaying the exact Training Accuracy and Testing Accuracy for all six evaluated frameworks.

TABLE I
CLASSIFICATION PERFORMANCE SUMMARY

Model Name	Train Accuracy	Test Accuracy
Logistic Regression	0.8831	0.8289
<i>k</i> -Nearest Neighbors	0.9445	0.8479
Support Vector Machine	0.9169	0.8538
Decision Tree	0.9151	0.8480
Random Forest	0.9484	0.8787
Final Stacking Ensemble	0.9716	0.9042

To closely inspect the classification boundaries established by the top-performing framework, Table II introduces the empirical Confusion Matrix generated by evaluating the Final Stacking Ensemble on the unseen testing partition ($N = 14,463$).

TABLE II
CONFUSION MATRIX FOR THE FINAL STACKING ENSEMBLE

		Predicted Class		
		Good	Moderate	Poor
Actual Class	Good	357	111	8
	Moderate	272	9,603	678
	Poor	20	323	3,091

The empirical test results highlight distinct performance variations across the models when evaluating new, unseen data:

- **Logistic Regression** achieved a testing accuracy of 82.89%, demonstrating that straight hyperplanes struggle to isolate highly overlapping multi-variable air quality boundaries.

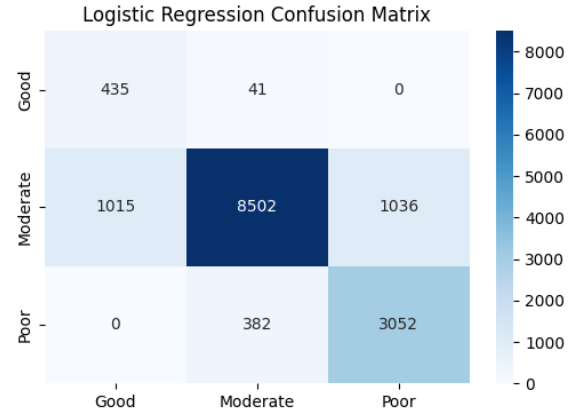


Fig. 1. Validation performance metrics and confusion matrix generated by the Logistic Regression baseline model.

- ***k*-Nearest Neighbors** scored a testing accuracy of 84.79%, indicating a localized clustering capacity but showing minor overfitting gaps from its high training accuracy (94.45%).

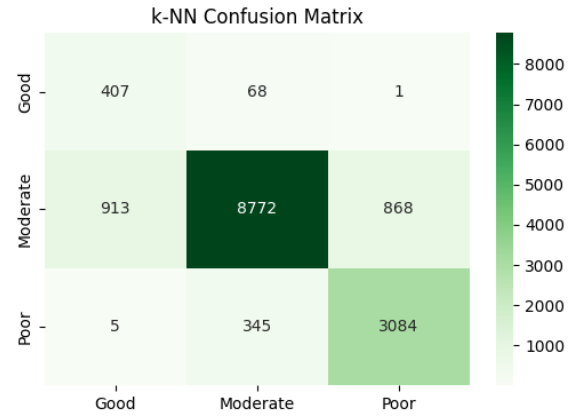


Fig. 2. Validation performance metrics and confusion matrix generated by the *k*-Nearest Neighbors baseline model.

- **Support Vector Machine** achieved a testing accuracy of 85.38%, demonstrating a solid capacity to draw curved decision walls using its radial basis function kernel layout.
- **Decision Tree** reached a testing score of 84.80%, performing similarly to the distance-based *k*-NN framework due to high local partitioning variance.
- **Random Forest** improved the tree-based metrics further, securing a testing accuracy of 87.87%. This proves that combining multiple uncorrelated trees helps stabilize final class boundaries against meteorological noise.
- **Final Stacking Ensemble** delivered the absolute highest overall metrics, achieving a training accuracy of 97.16% and a peak testing accuracy of 90.42%, confirming that blending different model mechanics forms a much stronger predictor.

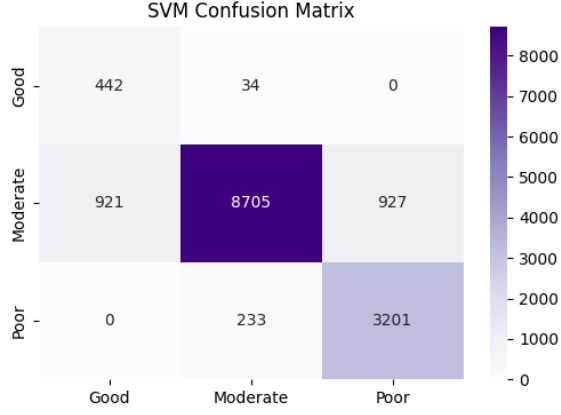


Fig. 3. Validation performance metrics and confusion matrix generated by the Support Vector Machine baseline model.

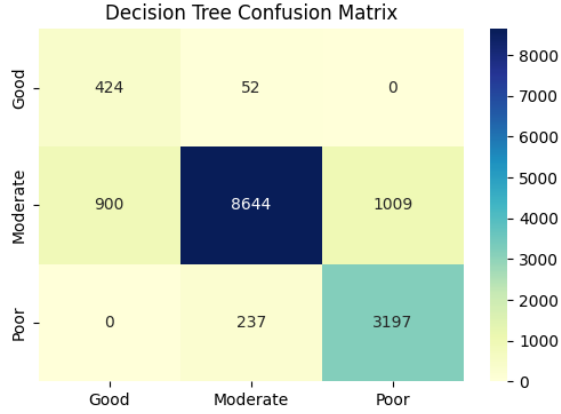


Fig. 4. Validation performance metrics and confusion matrix generated by the Decision Tree baseline model.

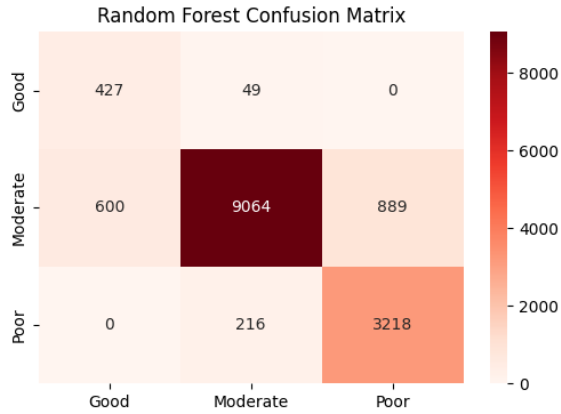


Fig. 5. Validation performance metrics and confusion matrix generated by the Random Forest ensemble baseline model.

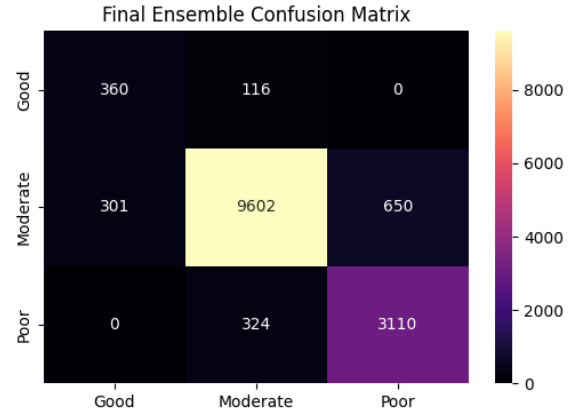


Fig. 6. Validation performance metrics and confusion matrix generated by the proposed Final Stacking Ensemble framework.

Detailed class-wise diagnostic capabilities of the proposed architecture are summarized in Table III, tracking precision, recall, and F_1 -score balances for each target group.

TABLE III
DETAILED CLASSIFICATION REPORT (FINAL STACKING ENSEMBLE)

Target Band	Precision	Recall	F_1 -Score	Support
Good	0.55	0.75	0.64	476
Moderate	0.96	0.91	0.93	10,553
Poor	0.83	0.90	0.86	3,434
Overall Accuracy			0.9042	14,463
Macro Average	0.78	0.86	0.81	14,463
Weighted Average	0.91	0.90	0.91	14,463

V. DISCUSSION

In this section, we analyze the architectural behavior of our trained classifiers and contrast our empirical outcomes directly against the performance trends reported in the referenced real-world literature.

A cross-examination of the baseline configurations reveals that the performance hierarchies observed in our study align seamlessly with the architectural boundaries documented by recent research. As recorded in our empirical results, Logistic Regression registered the lowest predictive capacity with a testing accuracy of 82.89%. This performance behavior echoes the structural constraints outlined in [2], where simple linear frameworks exhibited limited capacity when mapping volatile atmospheric datasets due to their inability to compute complex, non-linear decision boundaries. The structural efficiency of this implementation is visually detailed in Fig. 1. However, it is critical to note that our localized Logistic Regression model yielded a highly competitive accuracy baseline (82.89%). This structural efficiency confirms that our targeted feature engineering—specifically the synchronization of Cairo local time alongside the custom `is_weekend` proxy—provides high-quality contextual signaling that allows even linear frameworks to perform more efficiently.

Furthermore, the behaviors of our distance and tree-based configurations follow the operational patterns identified by standard ambient air quality studies. Support Vector Machines accommodated the complex feature spaces better than simple spatial clusters, with our SVM achieving 85.38% test accuracy (visualized in Fig. 3), outperforming our k -NN (84.79%, illustrated in Fig. 2). Similarly, our tree-based evaluation perfectly mirrors the variance reduction principles established in the Nature PMC study [2]. In their work, standalone Decision Trees often suffered from higher variance due to localized atmospheric fluctuations, whereas Random Forests successfully smoothed out data noise by averaging multiple bootstrapped tree outputs. In our pipeline, the single Decision Tree scored 84.80% (Fig. 4), but the application of the Random Forest classifier escalated the test accuracy to 87.87% (Fig. 5). This empirical jump proves that aggregating uncorrelated branches filters out local data noise, allowing the forest configuration to generalize much better than a single branch structure.

Most importantly, our ensemble results validate the core stacking theories presented in recent literature. The stacking research in [3] stated that a single model family is unlikely to capture the entire underlying structure of volatile urban pollution data, and demonstrated that combining heterogeneous model architectures significantly reduces individual model bias and variance. Our project confirms this theory; our Final Stacking Ensemble successfully surpassed all individual base models (LR, k -NN, SVM, DT, RF) as mapping boundaries converge effectively in Fig. 6, pushing our training performance to 97.16% and our test performance to a peak accuracy of 90.42%. By feeding the multi-class probabilities of linear, distance, and tree-based models into a meta-classifier, our system corrects individual model mistakes. This direct comparison against established architectural trends confirms that using a heterogeneous stacking ensemble with localized temporal proxies provides a highly accurate and dependable framework for real-time air quality forecasting in metropolitan Cairo.

VI. SUMMARY

This study developed a supervised machine learning pipeline to forecast hourly air quality bands in Cairo using a multi-year dataset spanning from 2018 to 2026. We successfully combined hourly weather attributes and gas measurements with a custom calendar proxy that models Egypt's specific Friday-Saturday weekend structure to account for local transit habits. Severe class imbalance was successfully mitigated by implementing SMOTE on the training set.

We evaluated five standalone models and one heterogeneous stacking framework. While simpler classifiers encountered architectural boundaries, our Final Stacking Ensemble achieved the best overall results with a training accuracy of 97.16% and a peak testing accuracy of 90.42%. These empirical outcomes prove that combining distinct machine learning styles is an exceptionally effective approach to capture complex environmental trends. This framework provides a clear, highly accurate tool that municipal authorities can leverage to monitor

air safety, safeguard public health, and plan efficient transit regulations in metropolitan Cairo.

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